MILITARY SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2.0 AMPERE

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

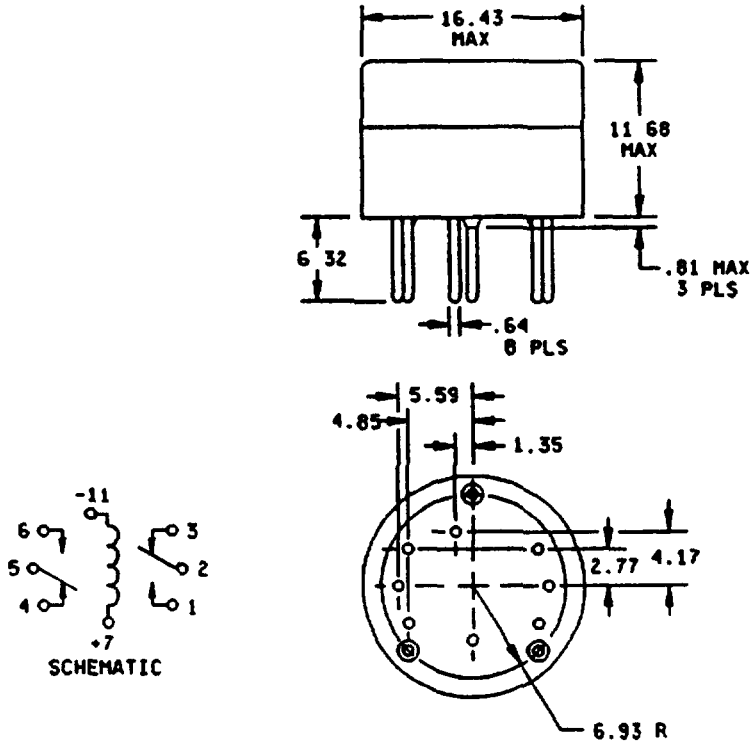


FIGURE 1 Dimensions and configuration

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mm	Inches	mm	Inches
0.64	.025	4.85	.191
0.81	.032	5.59	.220
1.35	.053	6.32	.249
2.77	.109	6.93	.273
4.17	.164	11.68	.460
		16.43	.647

NOTES:
1. Dimensions are in millimeters
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±0.25 mm (.010 inch)
4. All leads shall be electrically insulated from the case.
5. Circuit diagram shown on part is the terminal view.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Socket pin terminals shall be gold plated in accordance with MIL-G-45204, type II, class I, over underplating nickel 1.25 to 3.75 µm (50×10^{-6} to 150×10^{-6} inch) thick.
8. Solder pin terminals shall be hot solder dipped.
9. The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS

CONTACT DATA:

Load ratings

High level (relay case grounded):

Resistive
2.0 ampere at 28 V dc.
250 mA at 115 V ac 60 and 400 Hz case not grounded.
100 mA at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.100 ohm maximum

High level:

During life Not more than 5 percent of open circuit voltage

After life: 0.200 ohm maximum.

Low level:

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

During life: 1 ohm maximum

After life: 0.200 ohm maximum.

Contact bounce: 2.2 ms maximum (applicable to failure rate level "L").

Contact stabilization time. 2.0 ms maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated resistive current. Not applicable to ac load ratings.

COIL DATA: See table I.

Operate time: 4.5 ms maximum over temperature range with rated coil voltage.

Release time: 7.5 ms maximum over temperature range from rated coil voltage

ELECTRICAL DATA

Insulation resistance: 10,000 megohms at 500 V dc.

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Dielectric withstanding voltage

Sea level:

Between case, frame or enclosure and all contacts both in the energized and deenergized positions	500 V rms (60Hz)
Between case, frame, or enclosure and coils	500 V rms (60Hz)
Between all contacts and coils	500 V rms (60Hz)
Between open contacts in the energized and deenergized positions	500 V rms (60Hz)
Between open contacts	500 V rms (60Hz)

Altitude:

Between all terminals to case	125 V rms (60Hz)

ENVIRONMENTAL DATA:

Temperature range. -65°C to +125°C

Vibration (sinusoidal) 1/. MIL-STD-202, method 204, test condition G Contact chatter shall not exceed 10 µs maximum for closed contacts, and 1 µs maximum closure for open contacts.

Vibration (random) 1/: MIL-STD-202, method 214, test condition IH. Contact chatter shall not exceed 10 µs maximum for closed contacts, and 1 µs maximum closure for open contacts. Applicable to qualification and group C.

Shock (specified pulse) 1/. MIL-STD-202, method 213, except 3,500 g's half-sine, 2.0 ms. Contact chatter shall not exceed 10 µs maximum for closed contacts, and 1 µs maximum closure for open contacts.

Magnetic interference: Applicable

Resistance to soldering heat. Applicable

Acceleration. Applicable.

Salt atmosphere (corrosion) In accordance with method 1041 of MIL-STD-750.

Moisture resistance: Moisture resistance test shall be performed in accordance with method 106 of MIL-STD-202 immediately following terminal strength in group III of qualification inspection and group II of group C inspection. The following details apply:

Initial measurements: Not applicable.
Polarization: Applicable
Loading voltage. Not applicable.
Final measurements. At high humidity and after high humidity not applicable. After drying period - Insulation resistance, dielectric withstanding voltage, static contact resistance and pickup/dropout voltage shall be within specification limits.

PHYSICAL DATA

Terminal strength, method 211 of MIL-STD-202.

Pull test	Test condition A, 2.2 kg (1 pound) pull
Bend test	Test condition C, 1 1 kg (0 5 pound) pull
Twist test	As specified in MIL-R-39016.

1/ Relays under test are hard wired and potted in wax.

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Solderability. Applicable

Dimensions and configuration See figure 1

Design and construction The contacts are contained in a hermetically sealed chamber which does not contain the coil The coil is located outside the hermetically sealed chamber and protected by encapsulation.

Weight. 8.0 grams (0.282 ounce) maximum.

Seal: Hermetic, (less than 1 X 10^{-9} cm^3/s Helium at contact chamber). Seal test shall be 100 percent inspection and shall be performed as an in-process test on the contact chamber only. Seal testing is not applicable to qualification and quality conformance inspections.

Minimum marking. Military Part or Identifying Number (PIN), "J" with the date code (example J9237), circuit diagram, manufacturers' name or source code

LIFE TEST REQUIREMENTS·

High level: 100,000 cycles per relay
Low level. 100,000 cycles plus 900,000 cycles mechanical life.

PIN: M39016/56- (plus dash number from table I and suffix letter designation failure rate level as described in footnote 2/ of table I).

TABLE I. Dash numbers and characteristics. 1/

Dash number 2/		Coil voltage (V dc) 3/		At +25°C 4/				Over temperature range 4/		
Socket pin	Solder pin	Rated	Max	Coil resis ohms ±10%	Spec. pickup value (V dc)	Spec. hold value (V dc)	Spec. dropout value (V dc)	Spec. pickup value (V dc)	Spec. hold value (V dc)	Spec. dropout value (V dc)
001	002	5.0	5.8	45	2.7	1.4	0.22	3.5	2.3	0.14
003	004	6.0	7.3	50	3.0	2.0	0.3	4.4	3.2	0 18
005	006	12.0	14.8	200	6.0	4.0	0.6	8.4	6.5	0.41
007	008	26 5	32.0	1000	13.0	8.0	1.3	18.0	13.0	0.89

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: -004L.

3/ CAUTION: The use any coil voltage less than the rated voltage will compromise the operation of the relay.

4/ Pick-up, hold, and dropout values as shown are for test purposes only and are not to be used as design criteria

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QUALIFICATION INSPECTION

Qualification inspection and sample size See table II

TABLE II Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/56-007	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/56-001 M39016/56-003 M39016/56-005 M39016/56-008	2 units each PIN qualification inspection table, group II 1 unit terminal strength and solderability

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/57 may be used in addition to MIL-R-39016/56 data Prior to retention of qualification, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan

Qualification inspection (reduced testing) and sample size· See table III
If the relays produced for MIL-R-39016/56 are similar in construction and design except for the diodes to the relays produced for MIL-R-39016/57 then reduced testing for qualification of MIL-R-39016/56 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/57

TABLE III Qualification inspection (reduced testing)

Examination or test
2 units each coil voltage
Group II of qualification inspection table
1 unsealed sample unit for internal examination

CONCLUDING MATERIAL

Custodians
Navy - EC
Army - ER
Air Force - 85
NASA - NA

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5945-0906-01)

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